

Nathan Roos

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I am an ML / applied maths master's student (MVA, Télécom Paris, 4/4 GPA) with 2 main-track publications (NeurIPS 2026, WACV 2026), seeking a 6-month research internship from April 2027 in ML for biology, chemistry, healthcare or physics.

Publications

- D. Rizdvanetskiy, **N. Roos**, and P. Lutsik, "Data-Driven Soft Labeling Scales DNA Read Classification to Whole-Body Cell-Type Deconvolution," **NeurIPS 2026 (Main track)**
- **N. Roos**, E. Iakovleva, A. Gjergji, V.P. Pastore, and E. Tartaglione, "How I Met Your Bias: Investigating Bias Amplification in Diffusion Models," **WACV 2026 (Main track)**

Professional Experience

ML Research Intern (computational epigenomics)

KU Leuven, Dept. of Oncology, Lab. of Computational Cancer Biology and Epigenomics

 March 2026 - Aug. 2026  Leuven, Belgium

- Designed a deep learning framework (Transformers, LSTM) for cell-type deconvolution of bulk DNA samples from read-level methylation calls; accepted at NeurIPS 2026 (cf publications)
- Developed a reference-free, read-level deconvolution method: a probabilistic generative model fitted with EM (C++ implementation, custom likelihood tempering) that infers cell-type methylation profiles and proportions from multiple bulk samples

Data Scientist Intern

Banque de France

 Sept. 2025 - Feb. 2026  Paris, France

- Unsupervised detection of anomalous credit ratings using ML methods
- Development of statistical tests and Python analysis tools for the statistical testing of credit ratings

Education

Master MVA (double degree)

ENS Paris-Saclay

 Sept. 2026 - Sept. 2027  Paris, France

Advanced courses in modern ML: geometric deep learning, optimal transport, stochastic calculus, graph and text representation learning, LLMs, and statistical/geometric data analysis

Engineering degree in Applied Maths / Computer science

Télécom Paris, IP Paris

 Sept. 2023 - Sept. 2027  Paris, France

- GPA: 4/4 (top of the class)
- Computer Science: Deep Learning (MLP, CNN, VAE, Transformer, etc.), Databases, Theory of computation,

Logic, Compilation, Networking, Programming (Python, C, C++, Typescript/Javascript)

- Applied mathematics: Stochastic Modeling and Calculus, Numerical Analysis, Optimization, Statistics

Preparatory classes MPI*

Lycée Clémenceau

 Sept. 2021 - June 2023  Nantes, France

- GPA: 4/4 (Top of the class)
- Placed 8th in the Mines-Ponts National Competitive Examination

Projects

Research in debiasing schemes for Deep Learning

One year research track at Télécom Paris

 Sept. 2024 - June 2025

- State-of-the-art review of debiasing for computer vision ([link](#))
- Discovered previously unknown impact of sampling hyper-parameters on the amplification of bias by diffusion models. Paper accepted at WACV 2026 (cf publications)

Design and deployment of NetOps solution for an ISP

Group semester project at Télécom Paris

 Feb. 2024 - June 2024

- Jury Prize 2024 (top of 50+ projects)
- Design and deployment of a system to automate the updating of internet box configurations for an Internet Service Provider

Optimization of work scheduling on a flow network

15 months research project (TIPE) for the Competitive examinations to enter France's top Engineering Schools

 Feb. 2021 - April 2022

- Goal: find the work schedule on a flow network that maximizes road traffic
- Design, implementation in OCaml, and testing of 4 polynomial complexity approximation algorithms
- Search for proof of NP-hardness

Skills

ML/Deep Learning Transformers, RNN, LSTM, Diffusion models, VAE, CNN

Statistics/Probabilistic methods stochastic modeling, EM algorithms, statistical testing

Programming Python, C, C++, SQL, Typescript/Javascript, OCaml

Frameworks and Infra PyTorch, TensorFlow, Git, Docker, CI/CD, Slurm (HPC)

Languages French (Native), English (C1), German (B1)